

UNIT 5 REVIEW - Solutions

1. State in simplified form (answer only), the derivative of the following:

a) $y = (e^{\ln x})^2$ $y = x^2 \Rightarrow y' = 2x$

b) $y = 2^{\cos x} + \ln(3e)$ $y' = 2^{\cos x} \ln 2 (-\sin x)$

c) $y = \frac{\cos 2x}{\sin 2x (\cos^2 x - \sin^2 x)}$

$y = \frac{\cancel{\cos 2x}}{\sin 2x (\cancel{\cos 2x})} = \frac{1}{\sin 2x} = \csc 2x \Rightarrow$

$y' = -2 \csc 2x \cot 2x$

d) $y = 3^{\log_3(\cos 6x)}$

$y = \cos 6x \Rightarrow y' = -6 \sin 6x$

2. Differentiate each of the following. Simplify fully.

(a) $y = 3x^2 \tan(2x)$

$\frac{dy}{dx} = 6x \tan(2x) + \sec^2(2x)(2)(3x^2)$

$\frac{dy}{dx} = 6x(\tan(2x) + x \sec^2(2x))$

(b) $f(x) = \cos(x^3)$

$f'(x) = -\sin(x^3)(3x^2)$

$f'(x) = -3x^2 \sin(x^3)$

(c) $s = \sin(\ln 7t)$

$\frac{ds}{dt} = \cos(\ln(7t)) \left(\frac{7}{7t} \right)$

$\frac{ds}{dt} = \frac{\cos(\ln(7t))}{t}$

(d) $y = e^x \ln x$

$\frac{dy}{dx} = e^x \ln(x) + \frac{1}{x} e^x$

$\frac{dy}{dx} = e^x \left(\ln(x) + \frac{1}{x} \right)$

3. Differentiate the following and leave in unsimplified form.

(a) $f(x) = \tan^4(3 \ln x^2)$

$f(x) = \tan^4(3 \ln x^2)$

$f(x) = (\tan(6 \ln x))^4$

$f'(x) = 4(\tan(6 \ln x))^3 (\sec^2(6 \ln x)) \left(6 \left(\frac{1}{x} \right) \right)$

(b) $f(x) = 2 \ln \sqrt{1 - x + x^2}$

$f(x) = 2 \ln(1 - x + x^2)^{\frac{1}{2}}$

$f(x) = 2 \cdot \frac{1}{2} \ln(1 - x + x^2)$

$\frac{dy}{dx} = \frac{-1 + 2x}{1 - x + x^2}$

(c) $s(x) = e^\pi \ln(\sin x + \cos x)$

$\frac{ds}{dx} = e^\pi \times \frac{\cos x - \sin x}{\sin x + \cos x}$

4. Find the equation of the tangent to $y = x \ln x$ which is parallel to $3x - y + 7 = 0$.

If the tangent is parallel to $3x - y + 7 = 0$, then the

tangent will have the same slope.



$$3x - y + 7 = 0$$

$$y = 3x + 7$$

$$m = 3$$

$$y = x \ln x$$

$$\frac{dy}{dx} = \ln x + x \left(\frac{1}{x} \right)$$

$$\frac{dy}{dx} = \ln x + 1$$

$$3 = \ln x + 1$$

$$2 = \ln x$$

$$x = e^2$$

$$\text{When } x = e^2, y = 2e^2$$

$$m = 3, (e^2, 2e^2), y - mx + b$$

$$2e^2 = 3e^2 + b$$

$$-e^2 = b$$

$$\therefore y = 3x - e^2$$

5. Determine all points of inflection of $y = e^x \sin x$ on the interval $0 \leq x \leq 2\pi$.

$$y = e^x \sin x$$

$$\frac{dy}{dx} = e^x \sin x + e^x \cos x$$

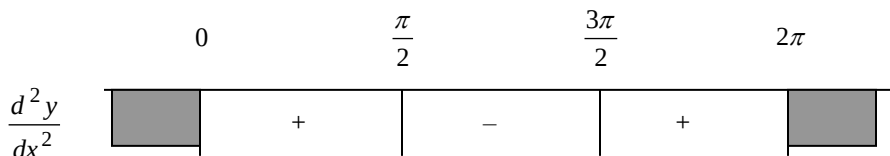
$$\frac{d^2y}{dx^2} = e^x \sin x + e^x \cos x + e^x \cos x - e^x \sin x$$

$$\frac{d^2y}{dx^2} = 2e^x \cos x$$

$$0 = 2e^x \cos x$$

$$2e^x = 0 \quad \cos x = 0$$

$$\text{No Solution} \quad x = \frac{\pi}{2}, x = \frac{3\pi}{2}$$



Therefore the points of inflection occur at $\left(\frac{\pi}{2}, e^{\frac{\pi}{2}} \right)$ and $\left(\frac{3\pi}{2}, e^{-\frac{3\pi}{2}} \right)$.

6. Determine the equation of the tangent to the curve $y = e^x + x$ which passes through the origin

$$\text{Point of tangency: } (x, e^x + x)$$

$$\text{slope of line passes through } (x, e^x + x) \text{ and } (0, 0): m = \frac{e^x + x}{x}$$

$$y' = e^x + 1$$

$$y' = m$$

$$e^x + 1 = \frac{e^x + x}{x} \Rightarrow xe^x + \cancel{x} = e^x + \cancel{x}$$

$$e^x(x - 1) = 0$$

$$x = 1$$

$$\text{Point of tangency: } (1, e + 1)$$

$$y - (e + 1) = (e + 1)(x - 1)$$

7. Determine any **Point(s) of inflection** of the function $f(x) = 2\sin^2 x - 1$ on the interval $-\pi \leq x \leq \pi$.

$$f'(x) = 4\sin x \cos x = 2\sin 2x$$

$$f''(x) = 4\cos 2x$$

$$\cos 2x = 0$$

$$2x = -\frac{3\pi}{2}, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}$$

$$x = -\frac{3\pi}{4}, -\frac{\pi}{4}, \frac{\pi}{4}, \frac{3\pi}{4}$$

	$-\pi$	$-\frac{3\pi}{4}$	$-\frac{\pi}{4}$	$\frac{\pi}{4}$	$\frac{3\pi}{4}$	π
$\frac{d^2y}{dx^2}$		+	-	+	-	+

8. A line with slope m passes through the origin and is tangent to $y = \ln\left(\frac{x}{3}\right)$. What is the value of m ?

$$\text{Point of tangency: } \left(x, \ln\left(\frac{x}{3}\right)\right)$$

$$\text{slope of line passes through } \left(x, \ln\left(\frac{x}{3}\right)\right) \text{ and } (0,0): m = \frac{\ln\left(\frac{x}{3}\right)}{x}$$

$$y' = \frac{\frac{1}{x}}{\frac{1}{3}} = \frac{1}{x}$$

$$y' = m$$

$$\frac{1}{x} = \frac{\ln\left(\frac{x}{3}\right)}{x} \Rightarrow \ln\left(\frac{x}{3}\right) = 1$$

$$\frac{x}{3} = e$$

$$x = 3e$$

$$m = \frac{1}{x} = \frac{1}{3e}$$

9. Find the values of a and b ($a \neq 0$) such that f is differentiable everywhere

$$(a) \quad f(x) = \begin{cases} \frac{\cos x}{\sin x + \cos x}, & x \leq 0 \\ ax + b, & x \geq 0 \end{cases}$$

$$(b) \quad f(x) = \begin{cases} \cos x, & x < \frac{\pi}{2} \\ ax + b, & x \geq \frac{\pi}{2} \end{cases}$$

f is differentiable everywhere, that means f is also continuous.

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = f(0)$$

$$\lim_{x \rightarrow 0^-} \frac{\cos x}{\sin x + \cos x} = 1$$

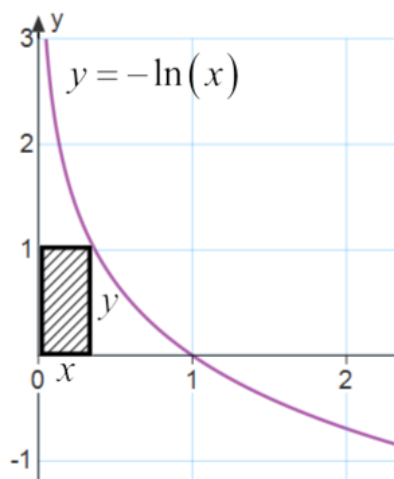
$$\lim_{x \rightarrow 0^+} (ax + b) = b \Rightarrow b = 1$$

$$f'(x) = \begin{cases} \frac{-\sin x (\sin x + \cos x) - (\cos x - \sin x) \cos x}{(\sin x + \cos x)^2}, & x \leq 0 \\ a, & x \geq 0 \end{cases}$$

$$f'_{-}(0) = f'_{+}(0)$$

$$-1 = a$$

10. A rectangle has its vertices on the x-axis, the y-axis, the origin, and the graph of $y = -\ln x$ in the first quadrant. Find the maximum possible area for such a rectangle.



$$A = xy, \quad y = -\ln x$$

$$A(x) = -x \ln x, \quad 0 \leq x \leq 1$$

$$A'(x) = -(\ln x + 1)$$

$$A'(x) = 0 \Rightarrow \ln x + 1 = 0$$

$$\ln x = -1$$

$$x = e^{-1}$$

$$A(e^{-1}) = -e^{-1} \ln(e^{-1})$$

$$= e^{-1}$$

11. Show that for any values A and B, $y = Ae^{2x} + Be^{-4x}$ satisfies the equation $y'' + 2y' - 8y = 0$.

$$y = Ae^{2x} + Be^{-4x}$$

$$y' = Ae^{2x}(2) + Be^{-4x}(-4)$$

$$y' = 2Ae^{2x} - 4Be^{-4x}$$

$$y'' = 2Ae^{2x}(2) - 4Be^{-4x}(-4)$$

$$y'' = 4Ae^{2x} + 16Be^{-4x}$$

$$LS = y'' + 2y' - 8y$$

$$LS = (4Ae^{2x} + 16Be^{-4x}) + 2(2Ae^{2x} - 4Be^{-4x}) - 8(Ae^{2x} + Be^{-4x})$$

$$LS = 4Ae^{2x} + 16Be^{-4x} + 4Ae^{2x} - 8Be^{-4x} - 8Ae^{2x} - 8Be^{-4x}$$

$$LS = 0$$

$$LS = RS$$

Therefore $y = Ae^{2x} + Be^{-4x}$ satisfies the equation $y'' + 2y' - 8y = 0$.